

Report — Proportional Control for Maneuvering Flapping Flight

Jean Pecquet

June 17, 2026

Contents

1	Summary	2
2	Model Description	2
2.1	Summary	2
2.2	Aerodynamic Force Derivation	2
2.3	Nondimensionalization	4
3	Two Analytical Cases	5
3.1	Stationary Flight	5
3.1.1	Wingbeat-averaged Aerodynamic Force	5
3.1.2	Power Efficiency	7
3.1.3	Force-Maximizing and Power-Minimizing Hover Kinematics	7
3.1.4	Implications for Morphology and Wingbeat Frequency	10
3.2	Stroke-Plane-Normal Flight	11
3.2.1	Rationale	11
3.2.2	Wingbeat-averaged Aerodynamic Force	11
3.2.3	Power-Optimal Trim	12
4	Controller Design	14
4.1	Hover Controller	14
4.1.1	Design Principle	14
4.1.2	Departure from Wingbeat-Averaged Picture	14
4.1.3	Proposed Control Scheme	15
4.1.4	Results	16
4.2	Generalized Controller	17
4.2.1	Design Principle	17
4.2.2	Proposed Control Scheme	17
4.2.3	Results	18

1 Summary

- A 2D point-mass model with quasi-steady blade-element wings reduces the wingbeat-averaged aerodynamic force to a product $F_a^* = q^* C_{F^*}$: a magnitude scale q^* set by flapping amplitude and frequency, and a force coefficient C_{F^*} set by the wing-pitch kinematics. The separation also holds in moving flight.
- Two control handles follow: the flapping amplitude ϕ_1 sets force magnitude and the mean pitch ψ_0 sets its direction. A proportional controller running an order of magnitude below the wingbeat frequency holds a stable hover across the full morphological parameter range.
- Within this quasi-steady model the inclined hover stroke plane carries no efficiency benefit; it is a force-steering choice, realized through a nonzero ψ_0 , rather than an energetic one.
- For flight along the stroke-plane normal (axial inflow) the cycle-mean force stays on the normal and the entire velocity dependence collapses onto the advance ratio J : the force coefficient falls with J and the force-maximizing pitch amplitude retunes downward.
- The maneuvering controller extends the hover controller by scheduling the stroke-plane angle to meet the inflow axially and slaving the pitch amplitude to J . In forward flight this keeps ψ_0 small and unsaturated at a near-flat power cost, far cheaper than holding the stroke plane horizontal.
- In unsteady closed-loop simulation the controller tracks accelerations, a pull-up, a fully reversing flight loop, and pure-pursuit prey capture; its binding limit is sustained, dynamic descent.

2 Model Description

2.1 Summary

We model a dragonfly-like insect with two pairs of wings. The sagittal plane is aligned with the fixed frame (x, z) plane, where z points opposite the direction of gravity. The wings in each pair beat symmetrically about the sagittal plane, and there is zero initial velocity or wind component normal to the sagittal plane, so the motion is constrained to the (x, z) plane, rendering the problem two-dimensional. A subsequent strong simplification is that the body is reduced to a point mass, so there is no coupling between body pitch and wing kinematics. In this manner we can focus entirely on the translational dynamics without concern for flight stability. The dynamical model is nondimensionalized using the body length L , the body mass m , and the gravitational time scale $\sqrt{L/g}$. The resulting dimensionless parameters, along with the wing kinematics angular parameters, are listed in [Table 1](#). Dimensionless quantities are denoted by an asterisk (*).

2.2 Aerodynamic Force Derivation

Each wing is modeled as a single, massless plate element of area A_w with its aerodynamic center at the $\frac{2}{3}$ -span station, $r = \frac{2}{3}L_w$, whose lift and drag coefficients are prescribed by functions of the instantaneous angle of attack α . These simplified functions, given by [Equation 5](#) and [Equation 6](#), nevertheless capture the essential phenomena. Parameters for the wing kinematics are defined in [Figure 1](#) and [Figure 2](#) for a single right-hand-side wing. Since the wings of each pair are assumed to beat symmetrically about the sagittal plane, the left-hand-side wing kinematics can be obtained by mirroring about the (x, z) plane.

Each wing element sweeps along an arc of radius $r = \frac{2}{3}L_w$ that lies inside the stroke plane, with normal vector $\vec{n} = \cos \gamma \vec{z} - \sin \gamma \vec{x}$, where γ is the stroke plane inclination angle relative to the horizontal. The sweep

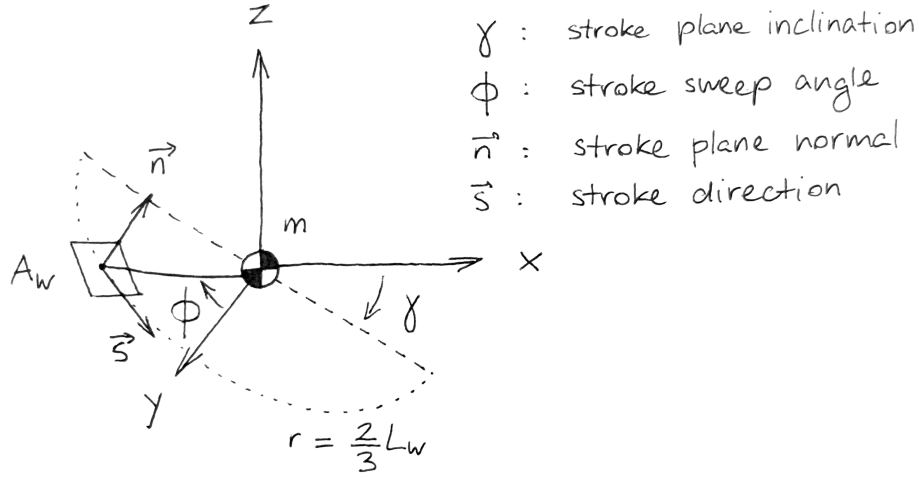


Figure 1: Wing stroke geometry

angle ϕ about $\vec{n}$ is defined relative to the side vector $\vec{y}$, positive counterclockwise about $\vec{n}$. At any point along the wing stroke, the wing velocity relative to the body is

$$\vec{v}_{w/b} = \frac{2}{3} L_w \dot{\phi} \vec{s} \quad (1)$$

with $\vec{s} = \cos \phi (\cos \gamma \vec{x} + \sin \gamma \vec{z}) - \sin \phi \vec{y}$. The sweep angle ϕ for wing pair i is prescribed by

$$\phi^{(i)}(t) = \phi_1 \sin(\omega t - \sigma_0^{(i)}) \quad (2)$$

The phase parameter $\sigma_0^{(i)}$ is set to 0 for the forewing pair, and to σ_0 for the hindwing pair. Both wing pairs beat at the same frequency ω and with the same amplitude ϕ_1 . In addition to the sweeping motion, the wing rotates about its own spanwise axis. We refer to this as the wing pitch. It is most easily described in the $(\vec{s}, \vec{n})$ plane. Figure 2 shows the pitch angle ψ defined relative to the stroke plane normal $\vec{n}$. ψ is prescribed for wing pair i by

$$\psi^{(i)}(t) = \psi_0 + \psi_1 \sin(\omega t - \delta_0 - \sigma_0^{(i)}) \quad (3)$$

Unlike for the sweep angle ϕ , the mean pitch angle ψ_0 may be nonzero, and can be used to redirect the mean aerodynamic force vector, as we will see.

It is convenient to define the chord direction vector $\vec{c} = \cos \psi \vec{n} - \sin \psi \vec{s}$. The angle of attack α is the angle between $\vec{c}$ and $\vec{v}_w$, the projection in the $(\vec{s}, \vec{n})$ plane of the wing velocity relative to the air. Care has to be taken as to the sign of α , as evidenced by the two panels in Figure 2. When $\vec{v}_w \cdot \vec{s} > \vec{c} \cdot \vec{s}$, α has the correct sign when defined from $\vec{v}_w$ to $\vec{c}$. When $\vec{v}_w \cdot \vec{s} < \vec{c} \cdot \vec{s}$, it has the correct sign when defined from $\vec{c}$ to $\vec{v}_w$. The following expression gives the correct sign:

$$\alpha = \sin^{-1} \left[\left(\frac{\vec{v}_w}{v_w} \times \vec{c} \right) \cdot (\vec{s} \times \vec{n}) \right] \quad (4)$$

Lift and drag coefficients are then obtained by

$$C_L(\alpha) = C_{L,0} \sin 2\alpha \quad (5)$$

$$C_D(\alpha) = C_{D,0} \cos^2 \alpha + C_{D,\frac{\pi}{2}} \sin^2 \alpha \quad (6)$$

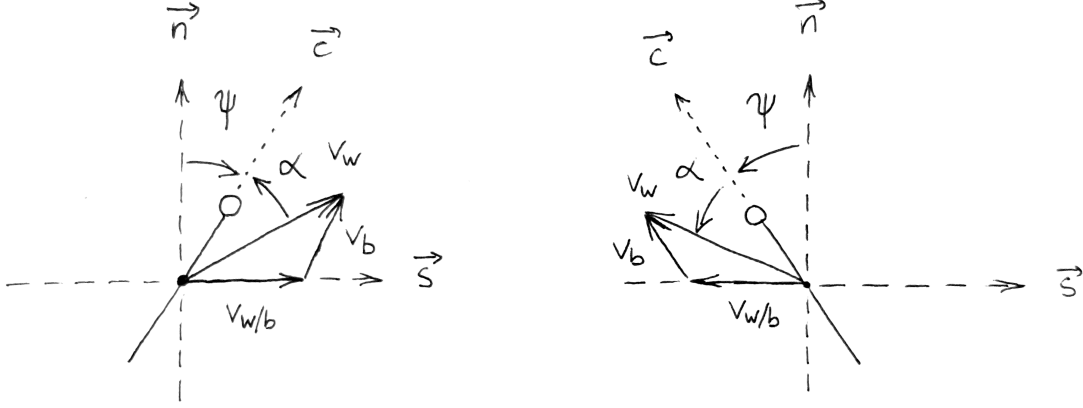


Figure 2: Wing pitch geometry and angle of attack for (a) $\vec{v}_w \cdot \vec{s} > \vec{c} \cdot \vec{s}$ and (b) $\vec{v}_w \cdot \vec{s} < \vec{c} \cdot \vec{s}$

with $C_{L,0} = 1.5$, $C_{D,0} = 0.1$, $C_{D,\frac{\pi}{2}} = 2$, as suggested by numerical simulations and experiments across the full range of angle of attack [1, 2]. The lift and drag vectors are, respectively,

$$\vec{d} = -\frac{\vec{v}_w}{v_w} \quad (7)$$

$$\vec{\ell} = \vec{d} \times (\vec{s} \times \vec{n}) \quad (8)$$

And the aerodynamic force of a single wing is

$$\vec{F}_{a,w} = \frac{1}{2} \rho_{\text{air}} A_w v_w^2 (C_D \vec{d} + C_L \vec{\ell}) \quad (9)$$

2.3 Nondimensionalization

The flying insect is subjected to gravity and the aerodynamic force on its wings. Body drag is neglected. The resulting equation of motion is

$$m (\ddot{x}\vec{x} + \ddot{z}\vec{z}) = \frac{1}{2} \rho_{\text{air}} A_w \sum_i^{N_w} \left[v_w^2 (C_D \vec{d} + C_L \vec{\ell}) \right]^{(i)} - mg\vec{z} \quad (10)$$

We nondimensionalize it using the body length L , the body mass m , and the gravitational time scale $\sqrt{L/g}$. The resulting dimensionless parameters, along with the already dimensionless wing angles, are summarized in Table 1. Ranges for the parameters L_w^* , A^*/m^* , and ω^* were obtained from the morphological data in [3] and the wingbeat frequency data in [4] for *Anisoptera*. The resulting dimensionless equation of motion is

$$\ddot{x}^* \vec{x} + \ddot{z}^* \vec{z} = \frac{1}{2} \frac{A^*}{m^*} \left(\frac{2}{3} L_w^* \phi_1 \omega^* \right)^2 \frac{1}{N_w} \sum_i^{N_w} \left[v_w^{*2} (C_D \vec{d} + C_L \vec{\ell}) \right]^{(i)} - 1 \vec{z} \quad (11)$$

Table 1: Dimensionless parameters

Parameter	Description	Range
$L_w^* = \frac{L_w}{L}$	Wing length ratio	[0.65, 0.85]
$A^*/m^* = \frac{N_w A_w \rho_{\text{air}} L}{m}$	Inverse wing loading	[0.2, 1.0]
$\omega^* = \omega \sqrt{\frac{L}{g}}$	Wing beat frequency	[8, 20]
γ	Wing stroke plane angle	[-180°, 180°]
ϕ_1	Wing flapping amplitude	[0°, 60°]
ψ_0	Mean wing pitch angle	[-90°, 90°]
ψ_1	Wing pitch amplitude	[0°, 90°]
δ_0	Wing pitch phase	[-180°, 180°]
σ_0	Fore/hind phase	[-180°, 180°]

3 Two Analytical Cases

3.1 Stationary Flight

3.1.1 Wingbeat-averaged Aerodynamic Force

In stationary flight, or *hovering*, the wingbeat-averaged aerodynamic force vector is opposed to and equal in magnitude to the weight vector. There are residual oscillations of the body due to fluctuations in the aerodynamic force over a wingbeat cycle, and inertial coupling between the body and wings (not modeled here). Assuming the corresponding body velocities are small compared to the characteristic wing velocity $\frac{2}{3}L_w^*\phi_1\omega^*$, we can approximate the mean aerodynamic force analytically by setting $\dot{x}^* = \dot{z}^* = 0$. As a result, the angle of attack is a simple function of the pitch angle ψ and the instantaneous sign of the flapping velocity:

$$\alpha = -\frac{\dot{\phi}}{|\dot{\phi}|} \left[\frac{\pi}{2} - \psi \right] \quad (12)$$

After making use of the appropriate trigonometric identities, this gives the lift and drag coefficients

$$C_L = -\frac{\dot{\phi}}{|\dot{\phi}|} C_{L,0} \sin 2\psi \quad (13)$$

$$\begin{aligned} C_D &= C_{D,0} \sin^2 \psi + C_{D,\frac{\pi}{2}} \cos^2 \psi \\ &= \frac{C_{D,\frac{\pi}{2}} + C_{D,0}}{2} + \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \cos 2\psi \end{aligned} \quad (14)$$

The aerodynamic force normal to the stroke plane (along $\vec{n}$) is pure lift, and the force within the stroke plane (along $\vec{s}$) is pure drag. Denoting these two components as $F_{w,n}^*$ and $F_{w,s}^*$, respectively, they are

$$F_{w,n}^* = \frac{1}{2} \frac{A_w^*}{m^*} \left(\frac{2}{3} L_w^* \dot{\phi} \right)^2 C_L \quad (15)$$

$$F_{w,s}^* = -\frac{\dot{\phi}}{|\dot{\phi}|} \frac{1}{2} \frac{A_w^*}{m^*} \left(\frac{2}{3} L_w^* \dot{\phi} \right)^2 C_D \quad (16)$$

At this point, it is useful to linearize, in a sense, about the $\phi = 0$ position, by treating $\vec{s}$ as a fixed vector, parallel to the (x, z) plane. This is a common modeling choice, see *e.g.* [5]. The effect of the simplification is that force is increasingly overestimated as the wingbeat amplitude ϕ_1 increases. The effect is small for

reasonable values of ϕ_1 . Note the numerical model retains the moving $\vec{s}$ (wing beats along an arc, not a straight line) and we only perform the simplification for the analytical derivation, so that the time integral depends on ϕ_1 only as a simple scaling factor. Note further that the simplification has no effect on the mean if the wing pitch is symmetric for both halves of the stroke, since $\langle F_s^* \rangle$ would be zero regardless.

With the above simplification made, we introduce the sweep kinematics $\phi(t^*) = \phi_1 \sin \omega^* t^*$, giving $\dot{\phi} = \phi_1 \omega^* \cos \omega^* t^*$. With the notation $\tau^* = \omega^* t^*$, we evaluate the wingbeat cycle integral $\langle \cdot \rangle = \frac{1}{2\pi} \int_0^{2\pi} (\cdot) d\tau^*$. An important question is how to deal with the sign flips due to the $\dot{\phi}/|\dot{\phi}|$ factors appearing in both F_n^* (through C_L) and F_s^* . We can actually fold them into the "velocity squared" factors as follows

$$\begin{aligned} \frac{\dot{\phi}}{|\dot{\phi}|} \dot{\phi}^2 &= \dot{\phi} |\dot{\phi}| \\ &= \phi_1^2 \omega^{*2} \cos \tau^* |\cos \tau^*| \end{aligned} \quad (17)$$

It is also convenient to introduce the wingbeat amplitude shorthand $s_0^* = \frac{2}{3} L_w^* \phi_1$. Evaluating the integrals, and multiplying by the number of wings (A_w^* becomes A^*), we obtain the total cycle-averaged aerodynamic force components

$$\begin{aligned} \langle F_n^* \rangle &= \frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \langle C_L |\cos \tau^*|^2 \rangle \\ &= -\frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 C_{L,0} \langle \sin 2\psi \cos \tau^* | \cos \tau^* | \rangle \end{aligned} \quad (18)$$

$$\begin{aligned} \langle F_s^* \rangle &= -\frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \langle C_D \cos \tau^* | \cos \tau^* | \rangle \\ &= -\frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{2} \langle \cos 2\psi \cos \tau^* | \cos \tau^* | \rangle \end{aligned} \quad (19)$$

Note the constant drag term vanishes because $\langle \cos \tau^* | \cos \tau^* | \rangle = 0$. The expressions feature a common $\frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^2$ factor, and a pitch- and aerodynamic-coefficient-dependent factor, which is lift-based for $\langle F_n^* \rangle$, and drag-based for $\langle F_s^* \rangle$. At this point, the only kinematics assumption we have made is sinusoidal flapping of the wings, while ψ remains generic. First, we can recover the analytical result of [5] for symmetric flapping with $\psi = -\frac{\pi}{4}$ for the $+\vec{s}$ stroke, and $\psi = +\frac{\pi}{4}$ for the $-\vec{s}$ stroke. Noting that $\cos \tau^* > 0$ for the $+\vec{s}$ stroke and < 0 for the $-\vec{s}$ stroke, and that $\langle \cos^2 \tau^* \rangle = \frac{1}{2}$, we recover the result

$$\langle F_n^* \rangle = \frac{1}{4} \frac{A^*}{m^*} (s_0^* \omega^*)^2 C_{L,0} \quad (20)$$

$$\langle F_s^* \rangle = 0 \quad (21)$$

We could obtain this result with our numerical model if we were to replace the sine wave with a square wave, and set $\psi_0 = 0$, $\psi_1 = \frac{\pi}{4}$. In reality, dragonfly wing pitch kinematics follow something between a sine wave and a square wave (*i.e.* a somewhat flattened sine wave). We stick to a sine wave here so as not to introduce higher harmonics ([6]) or a "flattening" coefficient ([7]), while keeping the more physical non-instantaneous stroke reversal (would be instantaneous with a square wave). To look at the effect of the pitch parameters on the force magnitude in the stroke plane and stroke-plane-normal directions, we introduce

$$q^* = \frac{1}{4} \frac{A^*}{m^*} (s_0^* \omega^*)^2 \quad (22)$$

$$C_{F^*,n} = -2C_{L,0} \langle \sin 2\psi \cos \tau^* | \cos \tau^* | \rangle \quad (23)$$

$$C_{F^*,s} = -\left(C_{D,\frac{\pi}{2}} - C_{D,0}\right) \langle \cos 2\psi \cos \tau^* | \cos \tau^* | \rangle \quad (24)$$

such that $\langle F_n^* \rangle = C_{F^*,n} q^*$ and $\langle F_s^* \rangle = C_{F^*,s} q^*$. It is also convenient to introduce the overall magnitude $C_{F^*} = \sqrt{C_{F^*,n}^2 + C_{F^*,s}^2}$.

3.1.2 Power Efficiency

In addition to the mean aerodynamic force, another quantity of interest is the mean power developed over a wingbeat. Because we assume the wings are massless, power is only due to the work of the aerodynamic force. In the hovering case, only the drag force does work, because the lift force is strictly normal to $\vec{s}$. It follows the (dimensionless) wingbeat-averaged power is

$$\begin{aligned}\langle P^* \rangle &= \langle |F_s^* s_0^* \omega^* \cos \tau^*| \rangle \\ &= \frac{1}{2} \frac{A^*}{m^*} (s_0^* \omega^*)^3 \langle C_D |\cos \tau^*|^3 \rangle\end{aligned}\quad (25)$$

Following the derivation of the *cost of endurance* in [2], we can normalize the above by introducing the quantity $u_{\text{ref}} = \sqrt{2mg/\rho A}$, more specifically its dimensionless form $u_{\text{ref}}^* = \sqrt{2/A^*/m^*}$, giving

$$\begin{aligned}\frac{\langle P^* \rangle}{u_{\text{ref}}^*} &= \frac{1}{2\sqrt{2}} \left[\frac{A^*}{m^*} (s_0^* \omega^*)^2 \right]^{3/2} \langle C_D |\cos \tau^*|^3 \rangle \\ &= 2\sqrt{2} (q^*)^{3/2} \langle C_D |\cos \tau^*|^3 \rangle\end{aligned}\quad (26)$$

When hovering, the wingbeat-average net aerodynamic force $C_{F^*} q^* = 1 \implies q^* = 1/C_{F^*}$. We introduce the hovering power coefficient

$$C_{P^*} = 2\sqrt{2} \frac{\langle C_D |\cos \tau^*|^3 \rangle}{C_{F^*}^{3/2}} \quad (27)$$

which is analogous to the steady value $C_D/C_L^{3/2}$. The velocity-weighted drag coefficient in the numerator plays the role of C_D , and the force coefficient C_{F^*} in the denominator, which is, in general, part lift and part drag, plays the role of C_L . To be more specific, C_{F^*} is lift-only as long as the wing stroke is symmetric ($\psi_0 = 0$), which, for hovering, requires the stroke plane to be horizontal. Hovering at an inclined stroke plane requires, as we will see in [subsubsection 3.1.3](#), $\psi_0 \neq 0$, resulting in drag contributing to C_{F^*} .

3.1.3 Force-Maximizing and Power-Minimizing Hover Kinematics

$C_L(\alpha)$ and $C_D(\alpha)$ being fixed, the force and power coefficients C_{F^*} and C_{P^*} defined in [subsubsection 3.1.1](#) and [subsubsection 3.1.2](#) respectively, depend only on the pitch parameters ψ_0 , ψ_1 , and δ_0 . We visualize these coefficients across two 2D slices, one at $\psi_0 = 0^\circ$, and one at $\delta_0 = 90^\circ$, in [Figure 3](#) and [Figure 4](#). The optimal values of each coefficient (maximum C_{F^*} and minimum C_{P^*}), indicated with white dots, both occur at $\psi_0 = 0^\circ$ and $\delta_0 = 90^\circ$, with different values of the pitching amplitude ψ_1 . Unsurprisingly, the C_{F^*} maximum occurs close to $\psi_1 = 45^\circ$, which maximizes lift, while the C_{P^*} minimum occurs close to $\psi_1 = 90^\circ$, which minimizes drag. To be exact, there are also maxima/minima at $\psi_0 = \pm 90^\circ$. They are interesting, but do not seem to correspond to typical dragonfly wingbeat patterns, so we disregard them here.

An important consequence of the maximum and minimum occurring at $\psi_0 = 0^\circ$ is that energy-efficient wing pitch occurs for a symmetric wingbeat, which results in a net mean force normal to the stroke plane, necessitating the use of a horizontal stroke plane. In other words, our quasi-steady lift and drag model is unable to account for the efficiency advantage of the dragonfly inclined stroke plane evidenced in other work [1, 2]. Since we want the model kinematics to still resemble dragonfly flight, we will impose a nonzero hover stroke plane angle γ (see [section 4](#)), independent of efficiency considerations for the present model. This will require the wingbeat-averaged force vector to be angled relative to the stroke plane normal by an amount $-\gamma$. As we will now see, this can be achieved with nonzero ψ_0 . Let us denote the angle of the force

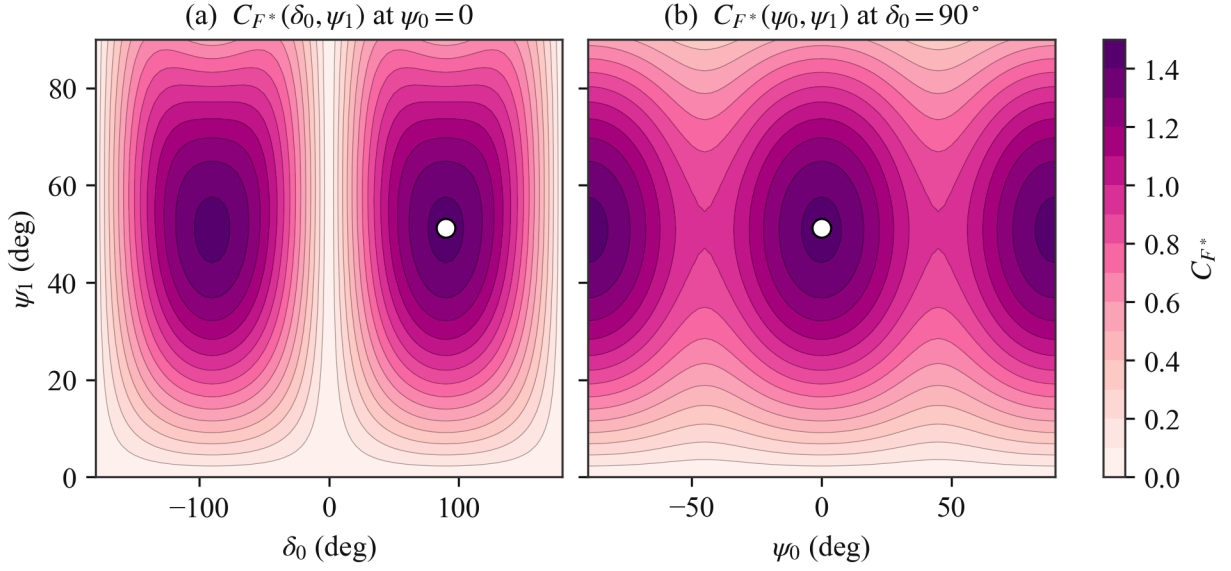


Figure 3: Dependence of C_{F^*} on pitch parameters at zero body velocity. Maximum shown as white dots.

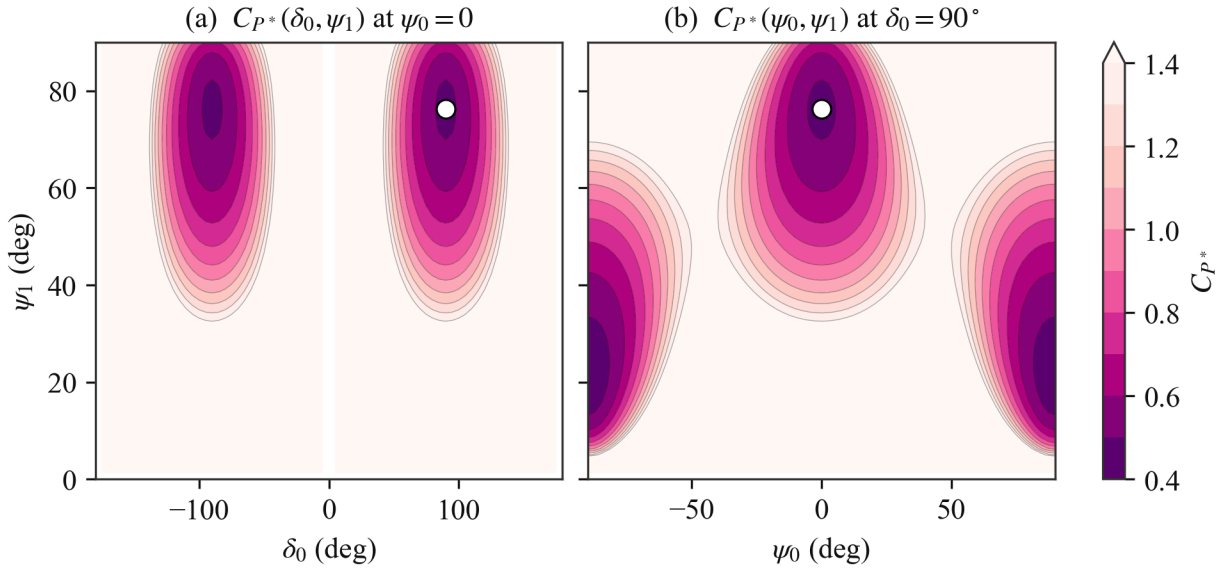


Figure 4: Dependence of C_{P^*} on pitch parameters for hovering. Minimum shown as white dots.

vector to the stroke plane normal $\vec{n}$ as β . Since the wingbeat-averaged force is $\langle \vec{F}^* \rangle = q^* (C_{F^*,n} \vec{n} + C_{F^*,s} \vec{s})$, β follows from the $C_{F^*,n}$ and $C_{F^*,s}$ coefficients (Equation 23 and Equation 24) as

$$\beta = -\text{atan2}(C_{F^*,s}, C_{F^*,n}), \quad (28)$$

taken positive counterclockwise from $\vec{n}$ (viewed with $\vec{x}$ pointing right and $\vec{z}$ up). Thus $\beta = 0$ for a force along $\vec{n}$, and $\beta = -\gamma$ aims the mean force along the vertical for hover at stroke-plane angle γ . Figure 5 maps the two components over the pitch plane. $C_{F^*,n}$ is largest for a near-symmetric stroke, whereas $C_{F^*,s}$ vanishes at $\psi_0 = 0$ and grows with $|\psi_0|$, supplying the off-normal force direction that an inclined-stroke hover requires.

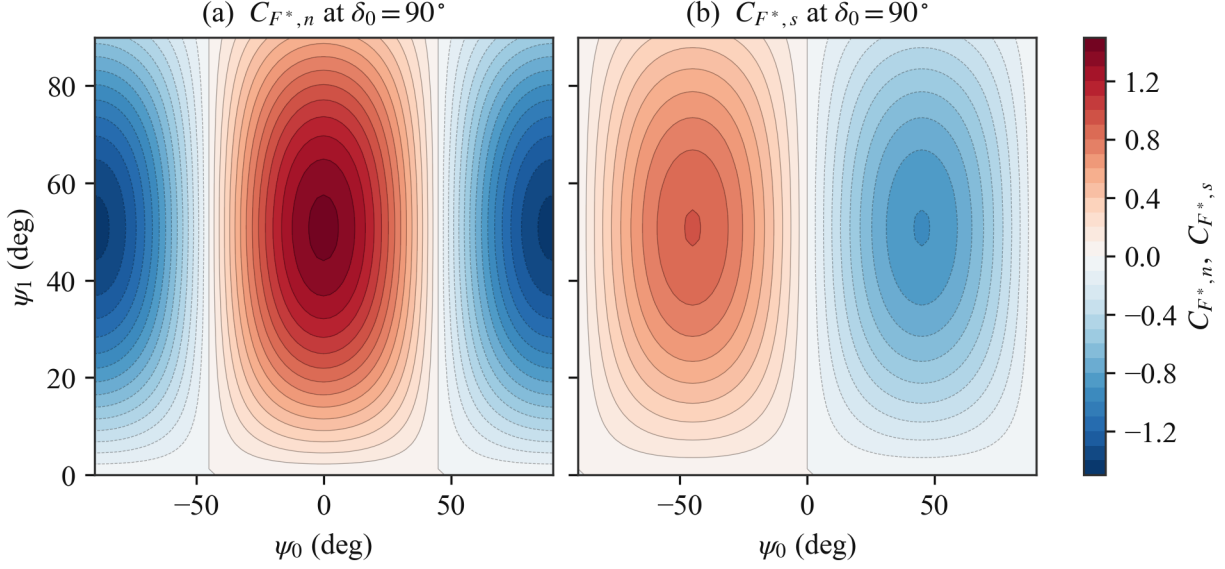


Figure 5: Stroke-normal (lift) component $C_{F^*,n}$ (a) and in-stroke (drag) component $C_{F^*,s}$ (b) of the force coefficient over the pitch plane at $\delta_0 = 90^\circ$. The force direction quadrant in the $(\vec{s}, \vec{n})$ plane can be inferred from the signs.

We further show that β depends on ψ_0 alone, independent of the pitch amplitude ψ_1 and the phase magnitude $|\delta_0|$. The weight $\cos \tau^* |\cos \tau^*|$ in Equation 23 and Equation 24 is antisymmetric under $\tau^* \rightarrow \tau^* + \pi$, a shift that also maps $\psi \rightarrow 2\psi_0 - \psi$ (since $\sin(\tau^* + \pi - \delta_0) = -\sin(\tau^* - \delta_0)$). Splitting each cycle average over the two half-strokes and applying the sum-to-product identities gives

$$\langle \sin 2\psi \cos \tau^* |\cos \tau^*| \rangle = \cos 2\psi_0 K, \quad (29)$$

$$\langle \cos 2\psi \cos \tau^* |\cos \tau^*| \rangle = -\sin 2\psi_0 K, \quad (30)$$

where the factor

$$K = \frac{1}{\pi} \int_0^\pi \sin[2\psi_1 \sin(\tau^* - \delta_0)] \cos \tau^* |\cos \tau^*| d\tau^* \quad (31)$$

contains the entire ψ_1 and $|\delta_0|$ dependence. It enters $C_{F^*,n}$ and $C_{F^*,s}$ as a common factor and so cancels in their ratio. Only its sign remains, fixed by the sign of δ_0 , flipping the force by 180° . With Equation 28 this yields the closed form

$$\beta = \text{atan2} \left[\left(C_{D, \frac{\pi}{2}} - C_{D,0} \right) \sin 2\psi_0, 2C_{L,0} \cos 2\psi_0 \right] \quad (32)$$

for $\delta_0 > 0$. In particular $\beta = 0, 90^\circ, 180^\circ$ at $\psi_0 = 0, 45^\circ, 90^\circ$, and near hover $\beta \approx \frac{C_{D,\frac{\pi}{2}} - C_{D,0}}{C_{L,0}} \psi_0 \approx 1.3 \psi_0$. The full numerical model retains the moving stroke tangent ($\vec{s}$ not constant) and departs from Equation 32 by at most $\sim 1^\circ$ over the ψ_1 and $|\delta_0|$ ranges, so the independence is exact in the analytical model and very nearly so in practice. We show $\beta(\psi_0)$ in Figure 6 along with the maximum C_{F^*} at a given ψ_0 .

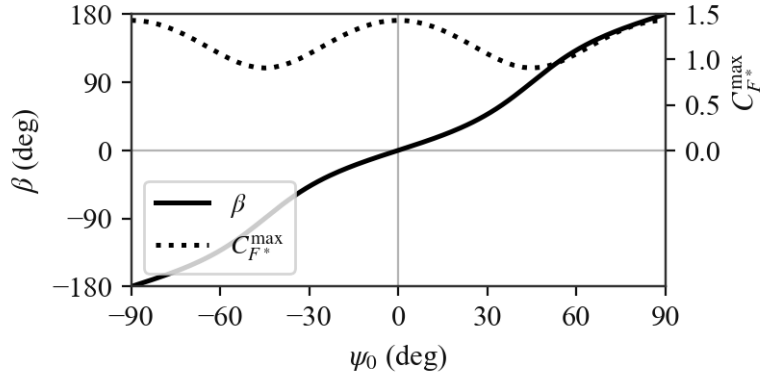


Figure 6: Effect of ψ_0 on force direction and wing stroke efficiency.

3.1.4 Implications for Morphology and Wingbeat Frequency

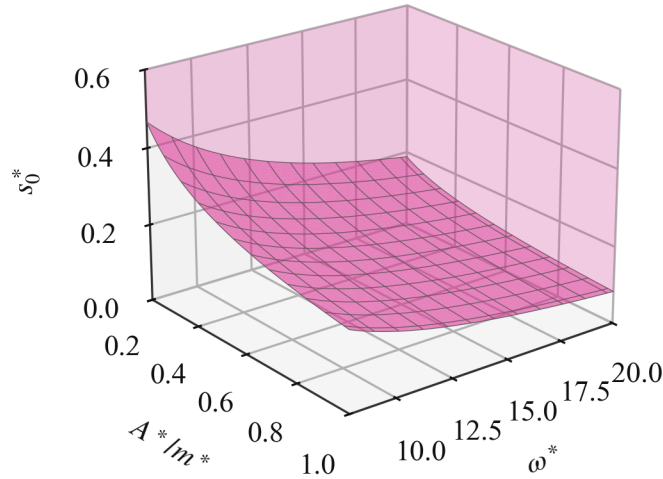


Figure 7: Approximate region of hover feasibility, with boundary surface.

Now that we have identified two families of “optimal” wing pitch kinematics, independent of morphology and wingbeat frequency, we take a brief look at what our findings imply about these parameters. Using the hovering criterion $F^* = 1$ for the maximum C_{F^*} pitch parameters, Figure 7 shows the required wingbeat

amplitude s_0^* (function of L_w^* and ϕ_1) for a given $(\frac{A^*}{m^*}, \omega^*)$. The sweep amplitude ϕ_1 can be recovered by

$$\phi_1 = \frac{3}{2} \frac{s_0^*}{L_w^*}$$

The very high amplitudes ($\phi_1 \approx 54^\circ$) associated with low values of both $\frac{A^*}{m^*}$ and ω^* suggest such combinations probably do not occur, and that the parameter space can be further restricted. The data source for ω^* does not report measurements necessary to obtain $\frac{A^*}{m^*}$, but we could maybe find other sources. We can estimate the typical hover wingbeat amplitude by assuming that the lowest ω^* occur at the highest $\frac{A^*}{m^*}$. For this set of parameters ($\omega^* = 8$ and $\frac{A^*}{m^*} = 1.0$) the amplitude is $s_0^* \approx 0.21$, which corresponds to $\phi_1 \approx 24^\circ$.

For the rest of the report, we use a reference configuration with intermediate values of all three parameters, yielding hover amplitudes close to the above estimate. The parameter values are shown in [Table 2](#).

Table 2: Reference configuration

A^*/m^*	L_w^*	ω^*	γ
0.6	0.75	14	40°

3.2 Stroke-Plane-Normal Flight

3.2.1 Rationale

Dragonflies have been observed to fly with their stroke plane normal $\vec{n}$ approximately aligned to the direction of travel [6]. We treat that aligned configuration here, with the body advancing along $\vec{n}$, and use the analysis for controller design in [subsection 4.2](#).

3.2.2 Wingbeat-averaged Aerodynamic Force

For flight along the stroke-plane normal the only velocity scale beyond the wing speed is the body speed v_b^* , which we carry through the advance ratio

$$J = \frac{v_b^*}{s_0^* \omega^*},$$

i.e. the body speed measured against the wing-speed scale $s_0^* \omega^*$. This choice has two virtues. First, the γ -invariance of the force magnitude carries over: $|\vec{F}_a^*|$ is unchanged when the stroke plane and the velocity vector are rotated together (verified numerically to machine precision), so a single map in J covers normal flight at every stroke-plane angle. Second, scaling the speed with $s_0^* \omega^*$ keeps every velocity in the problem proportional to the wing speed, which preserves the q^* magnitude scaling of [subsubsection 3.1.1](#): the body motion can only enter through the coefficient, which becomes $C_{F^*}(\psi_0, \psi_1, \delta_0; J)$.

This presumes the separation $F_a^* = q^* C_{F^*}$ of [subsubsection 3.1.1](#) survives at nonzero velocity. It does, exactly, for flight along the normal: the body velocity has no component along the in-plane sweep direction $\vec{e}_\parallel$, so the sweep term in v_s is untouched by the body motion,

$$v_s = \frac{2}{3} L_w \dot{\phi}, \quad v_n = \text{const}, \quad (33)$$

and the cycle geometry depends on the wingbeat amplitude only through $s_0^* \omega^*$. (Evaluating the scatter of F_a^*/q^* over random draws of $(\frac{A^*}{m^*}, \omega^*, \phi_1, L_w^*)$ at fixed pitch and fixed J returns a residual that is exactly

zero.) The wing therefore sees an oscillating sweep superposed on a uniform normal inflow—the classical heaving-plus-inflow kinematics of a blade element—and the angle of attack splits additively into the pitch and the relative-wind angle,

$$\alpha(\tau) = \psi(\tau) + \theta(\tau), \quad \theta(\tau) = \text{atan2}(v_s, v_n), \quad (34)$$

with $\tau = \omega^* t^*$. As the inflow vanishes ($J \rightarrow 0$) the relative-wind angle returns to $\theta \rightarrow \pm \frac{\pi}{2}$ and the hover result of [subsubsection 3.1.1](#) is recovered.

Writing the sweep-speed amplitude as $u = s_0^* \omega^*$ and the normal inflow through J , we have $v_s = u \cos \tau$, $v_n = Ju$ for a body advancing along the cycle-mean force, and $v_w = u \sqrt{\cos^2 \tau + J^2}$. Substituting into the lift and drag of [Equation 5](#) and [Equation 6](#) and cycle-averaging in the small- ϕ limit—where, as in [subsubsection 3.1.1](#), the stroke basis is frozen and the average of the vector equals the vector of the averages—gives the moving-flight coefficient

$$C_{F^*}(\psi_0, \psi_1, \delta_0; J) = 2 \sqrt{\langle \tilde{F}_s \rangle^2 + \langle \tilde{F}_n \rangle^2}, \quad (35)$$

$$\tilde{F}_n = \sqrt{\cos^2 \tau + J^2} (C_{L,0} \sin 2\alpha \cos \tau - J C_D), \quad (36)$$

$$\tilde{F}_s = -\sqrt{\cos^2 \tau + J^2} (C_D \cos \tau + J C_{L,0} \sin 2\alpha), \quad (37)$$

where $\tilde{F}_s, \tilde{F}_n$ are the stroke-frame components in units of $\frac{1}{2} \rho_{\text{air}} A_w u^2$ and α is given by [Equation 34](#). The entire velocity dependence has collapsed onto the single parameter J .

A parity argument sharpens the picture, paralleling the symmetric-pitch result of [subsubsection 3.1.1](#). For symmetric pitch ($\psi_0 = 0, \delta_0 = \frac{\pi}{2}$) the substitution $\tau \rightarrow \pi - \tau$ sends $v_s \rightarrow -v_s, \psi \rightarrow -\psi$, and hence $\alpha \rightarrow -\alpha$ at fixed v_n , so the $\tilde{F}_s$ integrand is odd and $\langle \tilde{F}_s \rangle = 0$: the cycle-mean force lies *exactly* along the stroke-plane normal at all advance ratios (verified numerically to 0.01°). The aligned wingbeat therefore produces a force purely along the flight axis; redirecting it—for instance to support weight away from vertical flight—requires breaking the symmetry through the mean pitch ψ_0 , the same lever used in hover. The coefficient then reduces to the single integral

$$C_{F^*}(J) = 2 \left| \left\langle \sqrt{\cos^2 \tau + J^2} (C_{L,0} \sin 2\alpha \cos \tau - J C_D) \right\rangle \right|, \quad (38)$$

which returns to the hover value as $J \rightarrow 0$ and decreases monotonically with J as the growing inflow detunes the angle of attack and bleeds off force—the wing losing authority as it advances into its own induced direction. At the hover-optimal pitch the coefficient falls from $C_{F^*} = 1.43$ at hover to 1.21 at $J = 0.2$ and 0.90 at $J = 0.4$ ([Figure 8](#)).

The pitch amplitude that maximizes C_{F^*} retunes with the inflow, decreasing from the hover optimum ($\psi_1 \approx 50^\circ$) toward $\psi_1 \approx 20^\circ$ at $J = 0.8$: the wing feathers less as the steady normal inflow takes over from the sweep in setting the angle of attack, which partly recovers the force lost at fixed pitch (to 1.17 at $J = 0.4$). We track the lift-producing optimum here; at large ψ_1 a second branch of nominally higher $|\tilde{F}_a^*|$ appears in which the feathered wing is carried backward by the inflow, with cycle-mean force opposed to $\vec{n}$, and is discarded as it cannot support weight. This $\psi_1^{\text{max}}(J)$ schedule is what the generalized controller of [subsection 4.2](#) slaves to.

3.2.3 Power-Optimal Trim

For straight flight along the normal at body velocity $\vec{v}_b$ we solve for the power-optimal sweep $\vec{u}$ producing weight-up, sweeping the redundant (γ, ψ_1) and trimming (ϕ_1, ψ_0) to the force at each speed. The vertical-flight result ([Figure 9\(a\)](#)) sets up the controller. The power-optimal stroke plane is *horizontal* ($\gamma = 0$) for hover and climb alike; only descent rewards a tilt. Climb stays feasible by re-tuning ψ_1 —the infeasibility of the hover

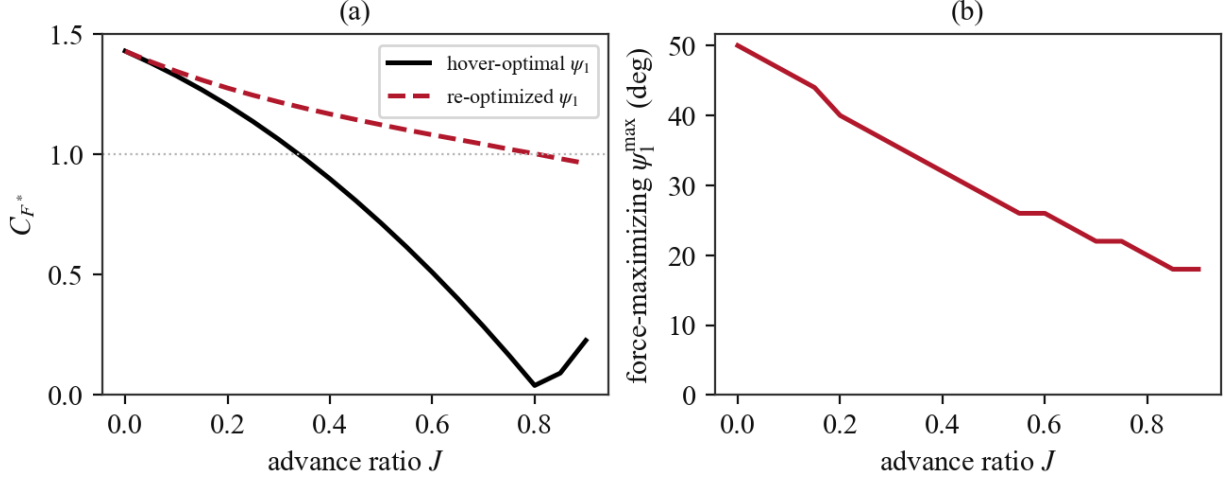


Figure 8: Axial inflow ($\chi = 0$), reference configuration. (a) Force coefficient $C_F^*(J)$ at the fixed hover-optimal pitch and at the re-optimized (lift-branch) pitch. (b) The force-maximizing amplitude $\psi_1^{\max}(J)$.

wingbeat is lifted once the pitch is allowed to adapt—at an aerodynamic cost that is mild until the climb work $\vec{v}_b \cdot \vec{F}_a^* = V_z W$ is added back to give the true metabolic cost; in descent the wing approaches an autorotative state, flapping little while the upward inflow carries the load. The off-normal forward-flight schedule of Figure 9(b)—where ψ_0 takes over the steering—belongs to the generalized controller of subsection 4.2.

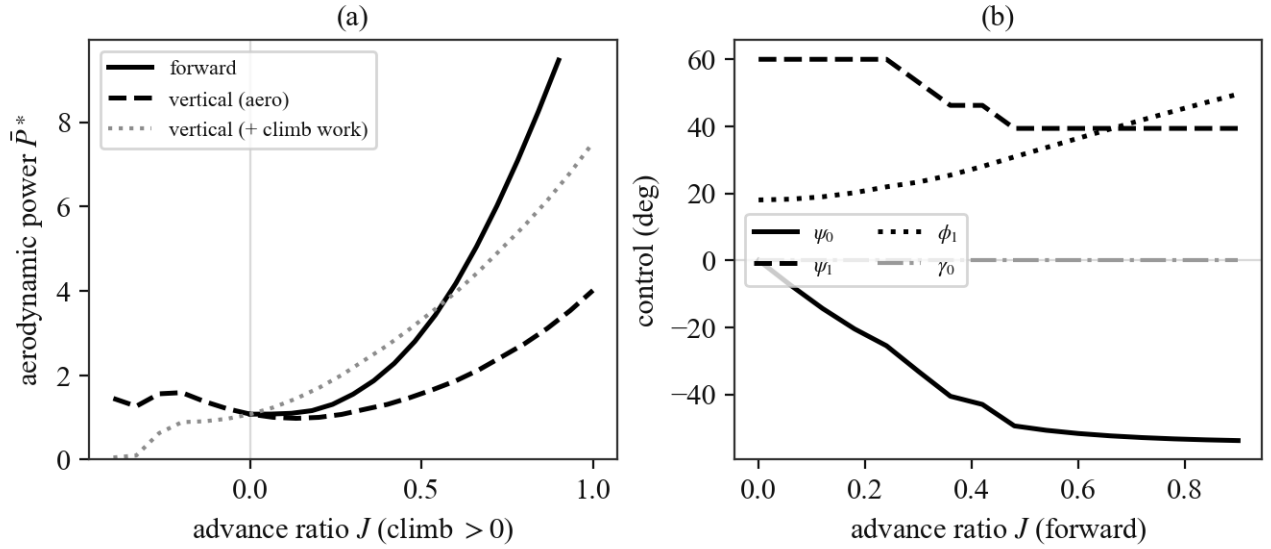


Figure 9: Power-optimal straight-flight trim vs. body velocity, reference configuration. (a) Aerodynamic power $\langle P^* \rangle$ vs. advance ratio for forward and vertical flight; the dotted curve adds the climb work $V_z W$ to the vertical aerodynamic power to give the metabolic cost. (b) The forward-flight control schedule: ψ_0 steers (and saturates), ψ_1 re-tunes down, ϕ_1 grows for magnitude, and the power-optimal γ stays at zero.

4 Controller Design

4.1 Hover Controller

4.1.1 Design Principle

Insights from [subsection 3.1](#) suggest a few possible ways to achieve hover control. Candidates for control variables can be split into two groups: those that control force magnitude (ϕ_1, ω^* through q^* and ψ_1, δ_0 through C_{F^*}), and those that control force direction (γ, ψ_0). Note this is not such a clean split, because ψ_0 also changes force magnitude through C_{F^*} , just more weakly than it changes direction. For simplicity, we would rather consider ω^* to be a fixed parameter, so we will exclude it at least for now. Since we also do not fully understand the factors that set the hovering stroke plane angle, we will also consider γ to be fixed for now (later on, we compare the effectiveness of γ and ψ_0 as directional control variables). In addition, it is beneficial to keep C_{F^*} at or near its maximum value, which sets ψ_1 and δ_0 . This leaves ϕ_1 (or s_0^*) for magnitude and ψ_0 for direction. At equilibrium, ψ_0 will be such that $\beta = -\gamma$ (force is vertical), and ϕ_1 such that $F_a^* = 1$. [Table 3](#) shows the required value of ψ_0 for different γ , for a reference configuration $L_w^* = 0.75$, $\frac{A^*}{m^*} = 0.6$, and $\omega^* = 14$. Different values of these parameters impose a different ϕ_1 for hover, which results in small changes in ψ_0 up to a few percent.

Table 3: Mean pitch for hover at various stroke plane angles for reference configuration

γ	ψ_0	$C_{F^*}^{\max}$
0°	0°	1.43
10°	-7.9°	1.40
20°	-15.1°	1.31
30°	-21.4°	1.21
40°	-26.7°	1.11

ϕ_1 is set by

$$\phi_1^{\text{hover}} = 3 \left(L_w^* \omega^* \sqrt{C_{F^*}^{\max}(\psi_0) \frac{A^*}{m^*}} \right)^{-1}$$

4.1.2 Departure from Wingbeat-Averaged Picture

Before designing a controller, we look at how the actual, unsteady hover dynamics differ from the wingbeat-averaged picture we have been working with so far. Indeed the body velocity will not be exactly zero throughout, and will oscillate according to the fluctuations of the aerodynamic force over a wingbeat cycle. There is an additional effect due to small but nonzero mass of the wings, which will make the body bob up and down in the direction of the flapping motion. We have neglected this in our work so far. In this part we will evaluate the relative importance of the two effects on the body oscillations.

A key parameter here is the fore/hind wing phase shift σ_0 . We expect the oscillations to be more pronounced when the wings beat in-phase ($\sigma_0 = 0^\circ$) than when they beat out-of-phase ($\sigma_0 = 180^\circ$). In fact, if the forewing and hindwing motions are identical except for the phase shift, the wing inertia effect will be zero (true for translation, not for body pitch, which we do not consider in the present model).

We introduce a relative wing mass parameter $m_w^* = m_w/m$. According to [\[3\]](#), values range from 0.01 to 0.03 for *Anisoptera*. We run one hover wingbeat to obtain the body oscillation peak-to-peak amplitude, comparing the ideal massless wing case to the $m_w^* = 0.02$ case, for the reference configuration of [Table 2](#) ([Figure 10](#)). The wing mass and aerodynamic contributions to the body oscillations appear to be of the same

order, except, as expected, when the wings beat out-of-phase. We will consider $m_w^* = 0$ in the following, but should keep in mind that this is only effectively true at $\sigma_0 = 180^\circ$.

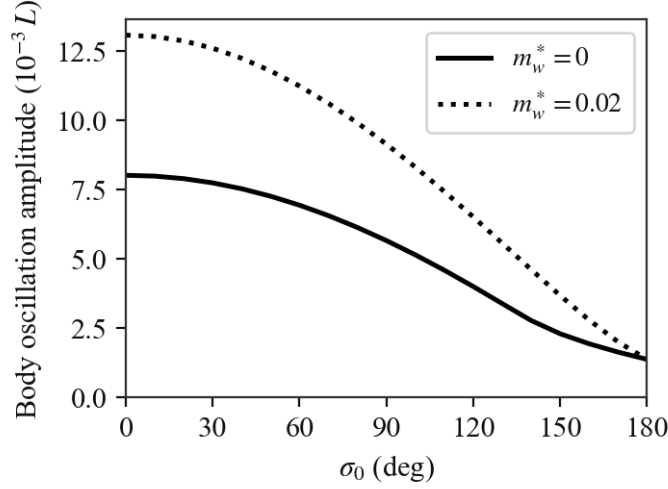


Figure 10: Effect of σ_0 and m_w^* on body oscillations during hover.

Another effect of the aerodynamic force oscillations is positional drift. For a constant, open-loop wingbeat pattern, the dragonfly settles in a steady-state that has a small but nonzero mean body velocity, when the equilibrium prediction is that the zero mean force condition occurs at exactly zero mean body velocity.

Overall these effects are relatively mild, with instantaneous body velocities of approximately $0.25s_0^*\omega^*$ in the worst case scenario ($\sigma_0 = 0^\circ$, $\omega^* = 8$, $A^*/m^* = 1.0$), and only about $\frac{1}{10}$ of that for $\sigma_0 = 180^\circ$, so the conclusions drawn from the equilibrium analysis should be applicable to controller design.

4.1.3 Proposed Control Scheme

Recall we identified ϕ_1 and ψ_0 as the two control variables for hover. ϕ_1 controls force magnitude, and ψ_0 controls force direction, with the secondary effect of reducing the force magnitude as it moves away from the stroke plane normal mean $\psi_0 = 0$. It seems natural to use ϕ_1 to control altitude, and ψ_0 to control fore-aft drift by tilting the force direction vector from the vertical. Specifically we propose the following control scheme

$$\begin{aligned}\psi_0 &= \psi_0^{\text{eq}} - K_x(x - x_0) \\ \phi_1 &= \phi_1^{\text{eq}} - K_z(z - z_0)\end{aligned}$$

Note that orienting the force in the $-x$ direction requires pitching up (increasing ψ_0), so we will need $K_x < 0$. To design the proportional gains for the reference configuration (Table 2) we begin by forming the Jacobian matrix of the wingbeat-averaged aerodynamic force $\vec{F}_a^*$ at the hover equilibrium, using finite differences, to evaluate the control effectiveness of the two variables about the force axes. The Jacobian is

$$J = \begin{bmatrix} \frac{\partial F_{a,x}^*}{\partial \psi_0} & \frac{\partial F_{a,x}^*}{\partial \phi_1} \\ \frac{\partial F_{a,z}^*}{\partial \psi_0} & \frac{\partial F_{a,z}^*}{\partial \phi_1} \end{bmatrix} \approx \begin{bmatrix} -2.06 & -0.05 \\ +0.93 & 5.83 \end{bmatrix}$$

Notably ϕ_1 has a negligible effect on force tilt from the vertical ($F_{a,x}^*$), while as we pointed out before ψ_0 does have a significant secondary effect on $F_{a,z}^*$. Since ϕ_1 cleanly drives $F_{a,z}^*$, its control loop will absorb the

ψ_0 cross term. The decoupled dynamics are (note mass matrix is I because of nondimensionalization)

$$\begin{aligned}\ddot{x}^* &= -2.06 \times \delta\psi_0 \\ \ddot{z}^* &= 5.83 \times \delta\phi_1\end{aligned}$$

So $\omega_n^2 = -2.06K_x = 5.83K_z$. We want ω_n slow enough that the controller sees the wingbeat-averaged force and not the fluctuations. Using $\omega_n = 1.5$ (10 times smaller than the reference wingbeat frequency $\omega^* = 14$) we get

$$\begin{aligned}K_x &= -1.09 \\ K_z &= 0.39\end{aligned}$$

4.1.4 Results

The step response is shown in Figure 11. The figure shows the reference configuration along with low and high wing loading configuration. A^*/m^* was chosen as the variable here because it had the greatest effect on the damping of the response. The response is very noticeably more damped as A^*/m^* increases. Over the range of parameters, using the same gains designed for the reference configuration, the achieved ω_n was close to the design value of 1.5, and the response did not vary very much except along the A^*/m^* axis. The controller was stable across the full parameter space.

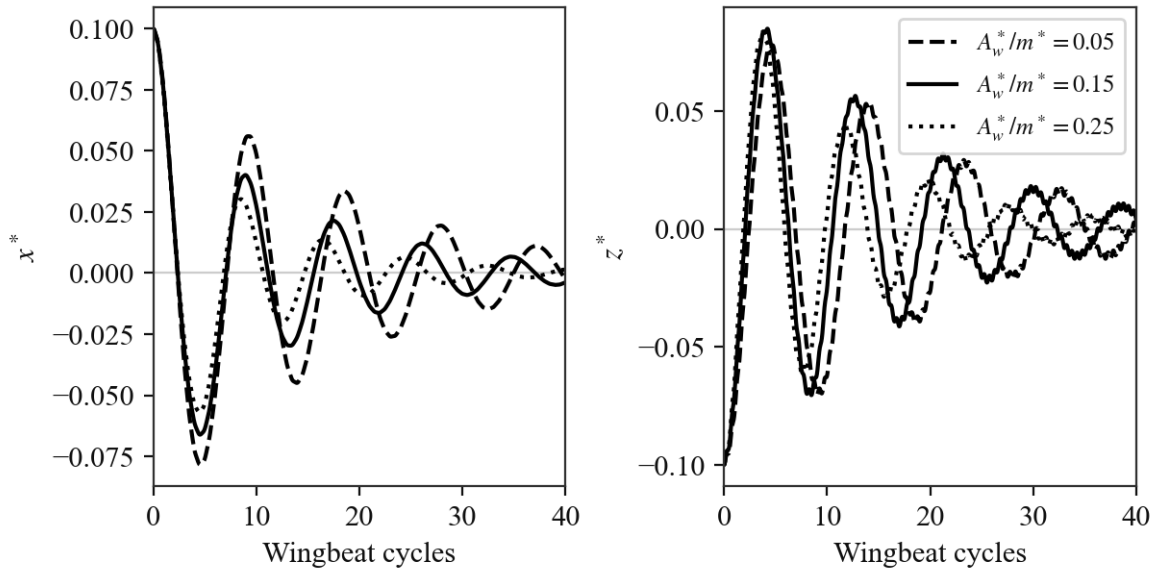


Figure 11: Effect of A^*/m^* on hover controller step response.

A more interesting result is the influence of stroke plane angle on damping, as shown in Figure 12. It appears that damping is strongest within the stroke plane, which means that vertical damping in the $\gamma = 0^\circ$ case is relatively low, and increases as the stroke plane is tilted. Horizontal damping decreases accordingly, with the end result being that there is a better match in the damping characteristics along both axes. Perhaps this is another advantage of hovering with an inclined stroke plane.

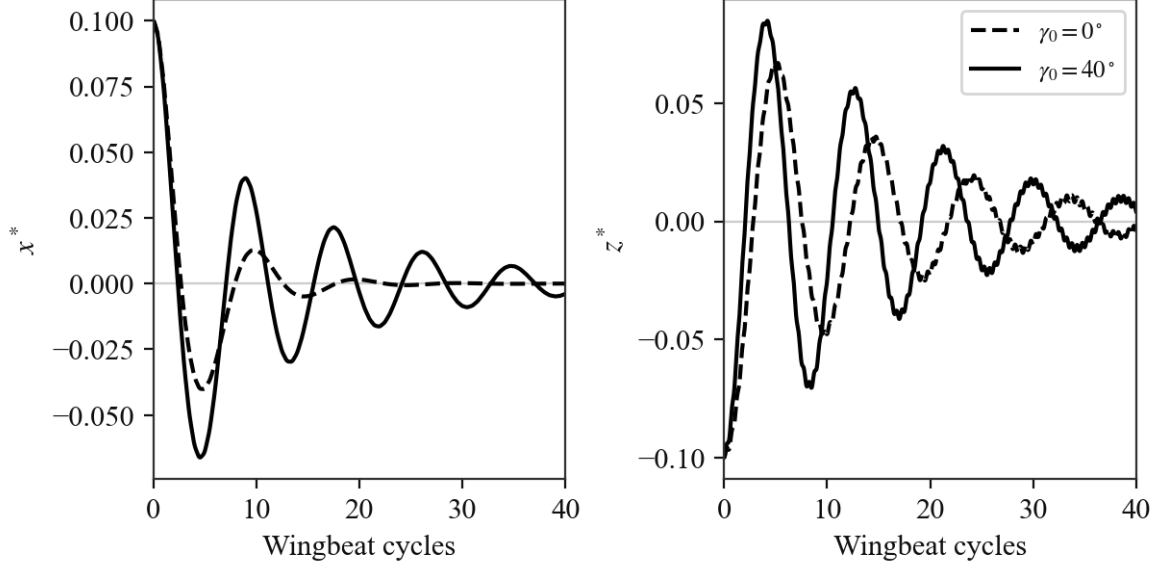


Figure 12: Effect of γ on hover controller step response.

4.2 Generalized Controller

4.2.1 Design Principle

Proposal to extend hover controller to moving/maneuvering flight

- Similar inner loop, with ϕ_1 controlling magnitude and ψ_0 controlling direction
- δ_0 still fixed at 90°
- ψ_1 becomes slaved to advance ratio, always giving maximum C_{F^*}
- γ becomes slaved to flight direction and advance ratio. At zero advance ratio, it is set to the hover value (fixed parameter, $\gamma_h = -40^\circ$). In the high limit, it is set to γ_f such that $\vec{n}$ points in the body velocity direction. In between, γ is set to $f(J)\gamma_f + (1 - f(J))\gamma_h$, with $f(J)$ between 0 and 1, $f'(J) \geq 0$, $f(0) = 0$, $f(\infty) = 1$.
- Because $\gamma_h \neq 0$, there is an asymmetry in this basic formulation between flight along $+x$ and along $-x$. We can resolve this by flipping the sign of γ_h depending on the desired direction of motion.

The controller should take into account the inclined hover stroke plane, and transition smoothly between and hovering and steady motion. The in-plane body velocity component should remain small compared to the wing velocity relative to the body.

4.2.2 Proposed Control Scheme

We concretize the principle above as a velocity-scheduled controller. The pitch phase is held at $\delta_0 = 90^\circ$. The stroke-plane angle follows

$$\gamma(J) = f(J) \gamma_f + (1 - f(J)) \gamma_h, \quad f(J) = 1 - e^{-(J/J_c)^2}, \quad (39)$$

where $\gamma_f = \text{atan2}(-\hat{v}_x, \hat{v}_z)$ aligns the stroke-plane normal $\vec{n}$ with the body velocity $\hat{v}$ (axial inflow, $\chi \rightarrow 0$), $\gamma_h = -40^\circ \text{ sign}(\hat{v}_x)$ is the inclined hover plane leaning toward the commanded direction, and the blend f is monotone with $f(0) = 0$, $f(\infty) = 1$ (we use $J_c = 0.35$). The pitch amplitude ψ_1 is slaved to the advance ratio at the force-maximizing value $\psi_1^{\max}(J)$ of [subsubsection 3.2.2](#) ([Figure 8](#)). The two hover handles then remain: ϕ_1 for magnitude and ψ_0 for direction, trimmed to the required force by the same 2×2 inversion as the hover inner loop, evaluated at the current body velocity. With no body drag, steady straight flight needs the weight-up force at every speed, so this schedule is the equilibrium trim and maneuvering is the transient excursion of (ϕ_1, ψ_0) away from it.

4.2.3 Results

We trim the proposed controller to weight-up along forward, climb, and descent velocity rays, and compare it to a baseline that holds the stroke plane horizontal ($\gamma = 0$) and re-optimizes ψ_1 for power.

In forward flight the schedule tilts the stroke plane toward vertical so the wing meets the inflow axially, and the cycle-mean force sits near the vertical with little redirection. The mean pitch ψ_0 *decreases* with speed, from its 27° hover value to 5° at $J = 1.0$, staying well clear of its bound; ϕ_1 stays small; and the aerodynamic power is nearly flat ($\langle P^* \rangle \approx 1.0\text{--}1.4$; [Figure 13](#)). The horizontal baseline meets the same flight as in-plane wind and holds the force vertical only by driving ψ_0 toward its limit (-57° at $J = 0.8$) and flapping harder, at a steeply rising cost ($\langle P^* \rangle \approx 7.4$ at $J = 0.8$, roughly five times the proposed controller).

In vertical flight the ranking reverses: climb and descent are already axial at $\gamma = 0$, so the baseline trims them at $\psi_0 = 0$ and low power while the proposed schedule carries the inclined hover plane into low-speed vertical flight at a modest power penalty ([Figure 13](#)). Forward flight and climb remain feasible beyond $J = 1$ for both controllers; descent limits the proposed controller to $J \approx 0.35$, where the over-supporting inflow exceeds the weight at the minimum flapping amplitude.

A closed-loop test of the scheme under the unsteady dynamics, with the control updated once per wingbeat, is shown in [Figure 14](#): a forward dash tracks an accelerate–cruise–decelerate command to $J \approx 0.5$ (peak velocity error 0.15), and a pull-up follows the velocity command through a quarter-turn from forward to climb (peak speed error 0.07).

A more demanding test reverses the body velocity in both axes: the controller is commanded around a closed circular flight path—forward, climb, backward, descent, and back to rest—so that v_x and v_z each change sign twice ([Figure 15](#)), tracked with a peak velocity error of 0.05. The loop is deliberately kept gentle (advance ratio $J \lesssim 0.16$): a faster loop fails at the bottom of the circle, where the body descends as v_x reverses, so the axial-descent geometry and the $\pm x$ stroke-plane sign flip coincide, ψ_0 saturates at its bound, and the body drops out of the loop—the controller’s limit in dynamic descent.

The same velocity tracker drives the prey-capture case studied in the project documentation. We close the loop with pure-pursuit guidance: at each control update the commanded velocity is aimed along the line of sight to the prey, at a fixed speed exceeding the prey’s. Starting from rest, the dragonfly closes the range monotonically and intercepts a horizontally moving target within $0.5 L$ ([Figure 16](#)). A descending target would instead inherit the dynamic-descent limit noted above.

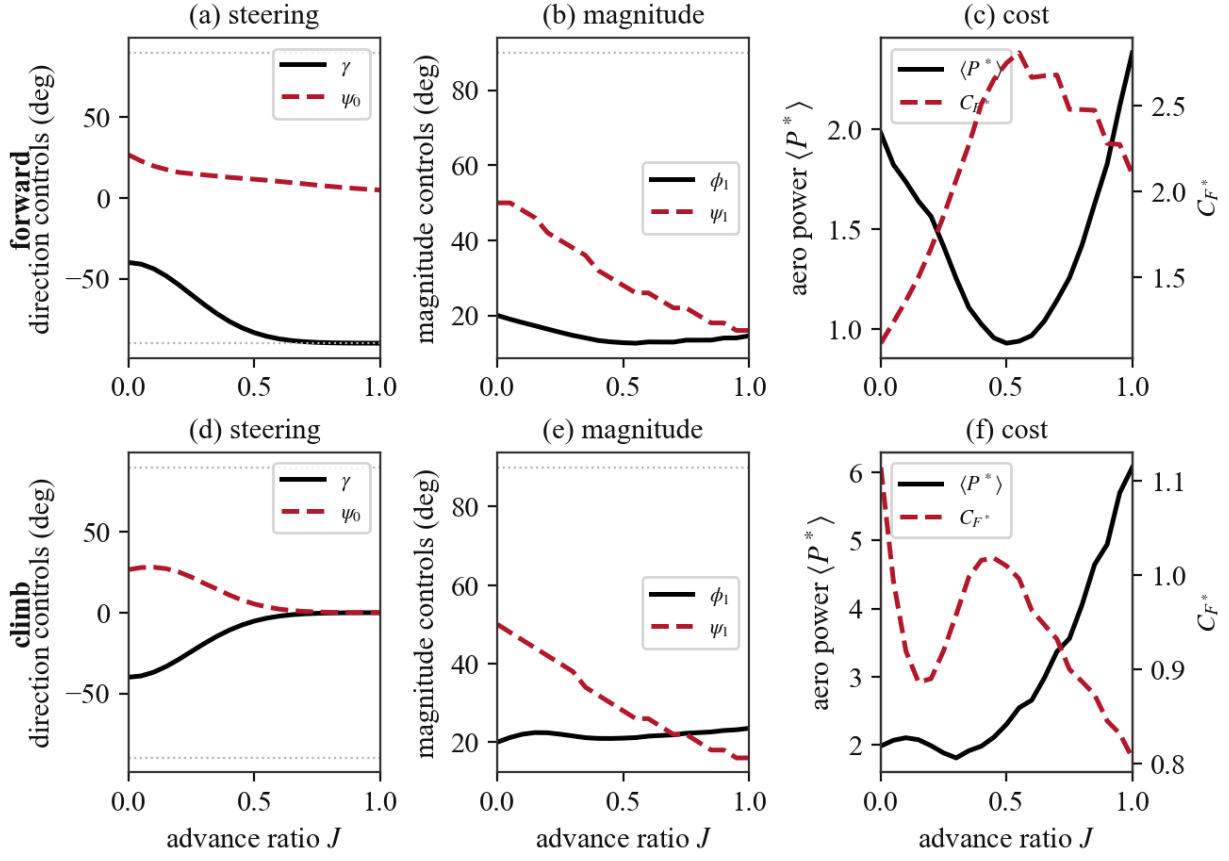


Figure 13: Proposed controller, reference configuration; top row forward flight, bottom row vertical climb. (a,d) Direction controls γ and ψ_0 ; (b,e) magnitude controls ϕ_1 and ψ_1 ; (c,f) aerodynamic power $\langle P^* \rangle$ (solid) and force coefficient C_F^* (dashed). Dotted lines mark the $\pm 90^\circ$ control bounds.

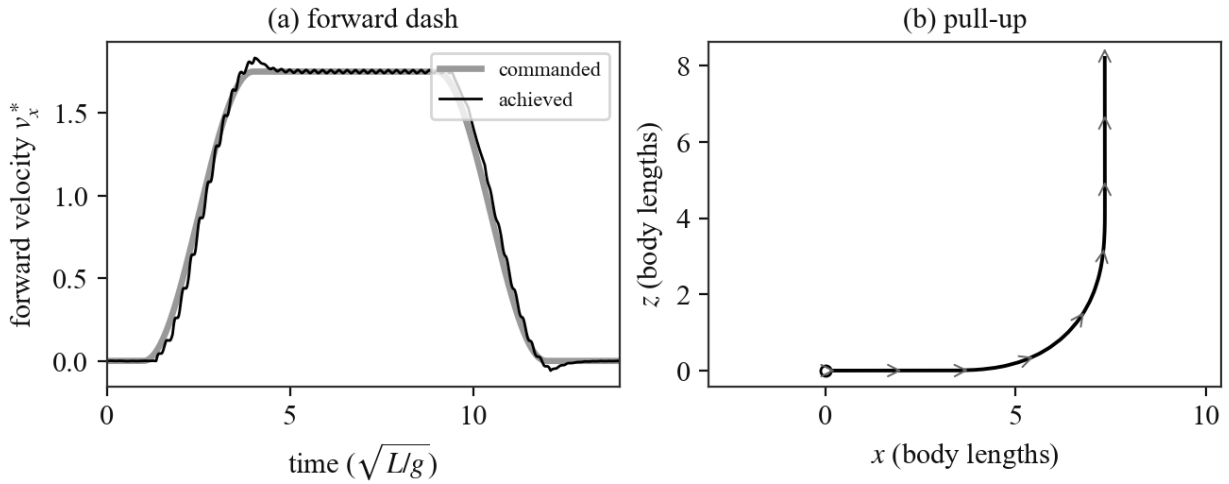


Figure 14: Closed-loop maneuvers under the proposed controller (unsteady simulation, control updated once per wingbeat). (a) Forward dash: commanded versus achieved forward velocity. (b) Pull-up: the sagittal-plane path, with body-velocity arrows rotating from horizontal to vertical.

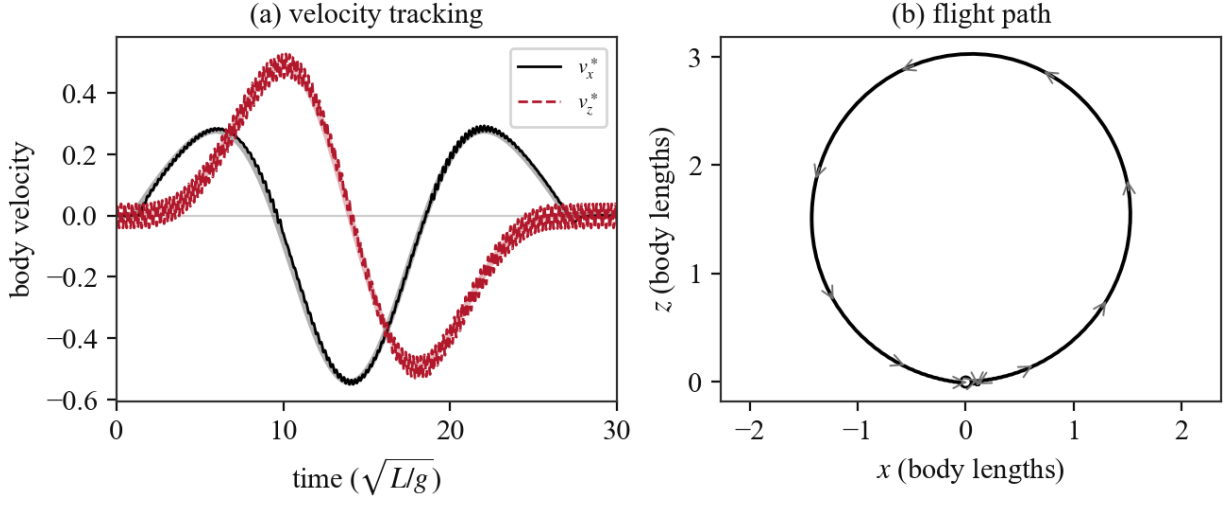


Figure 15: Closed-loop circular maneuver under the proposed controller (unsteady simulation). (a) Body velocity components v_x^* and v_z^* , commanded (light) versus achieved; both reverse sign twice. (b) The sagittal-plane flight path with body-velocity arrows.

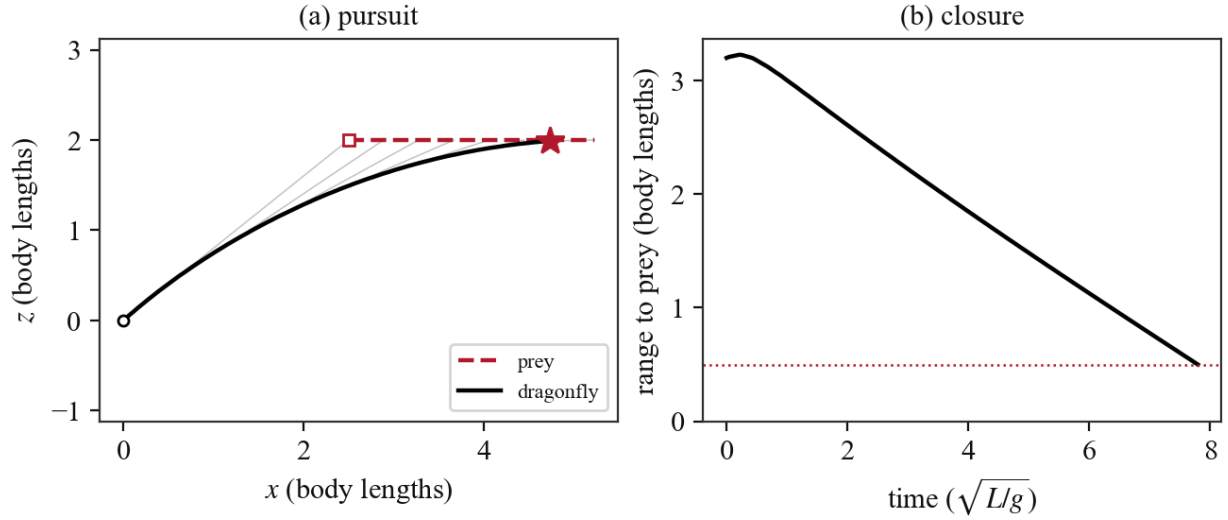


Figure 16: Pure-pursuit prey capture under the proposed controller. (a) Sagittal-plane paths of the dragonfly (from rest at the origin) and the prey (moving horizontally), with line-of-sight rays; the star marks interception at range $0.5 L$. (b) Range to the prey versus time.

References

- [1] Z. Jane Wang. The role of drag in insect hovering. *Journal of Experimental Biology*, 207(23):4147–4155, 11 2004. ISSN 0022-0949. doi: 10.1242/jeb.01239. URL <https://doi.org/10.1242/jeb.01239>.
- [2] Z. Jane Wang. Aerodynamic efficiency of flapping flight: analysis of a two-stroke model. *Journal of Experimental Biology*, 211(2):234–238, 01 2008. ISSN 0022-0949. doi: 10.1242/jeb.013797. URL <https://doi.org/10.1242/jeb.013797>.
- [3] J. M. Wakeling. Odonatan wing and body morphologies. *Odonatologica*, 26(1):35–52, 1997. URL <https://natuurtijdschriften.nl/pub/592186>.
- [4] C. Aracheloff, R. Garrouste, A. Nel, R. Godoy-Diana, and B. Thiria. Subtle frequency matching reveals resonant phenomenon in the flight of Odonata. *Journal of The Royal Society Interface*, 21(219), 2024. doi: 10.1098/rsif.2024.0401.
- [5] Owen C. Wetherbee and Z. Jane Wang. Stable flapping flight in morphological space: Model, simulation, and explicit stability criteria. *Proceedings of the National Academy of Sciences*, 123(18):e2533138123, 2026. doi: 10.1073/pnas.2533138123. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2533138123>.
- [6] Akira Azuma, Soichi Azuma, Isao Watanabe, and Toyohiko Furuta. Flight mechanics of a dragonfly. *Journal of Experimental Biology*, 116:79–107, 1997.
- [7] Gordon J. Berman and Z. Jane Wang. Energy-minimizing kinematics in hovering insect flight. *Journal of Fluid Mechanics*, 582:153–168, 2007.